

Organise the source

Move these compressed files to the `/usr/src` directory and uncompress them (as *root*, but you already knew that!), as follows:

```
mv linux-2.4.21.tar.bz2 /usr/src
mv rtlinux-3.2-wr.tar.bz2 /usr/src
cd /usr/src
tar -jxvf linux-2.4.21.tar.bz2
tar -jxvf rtlinux-3.2-wr.tar.bz2
```

Let's give these extracted directories shorter, easier-to-use aliases by creating soft links to them:

```
ln -s linux-2.4.21 linux
ln -s rtlinux-3.2-wr rtlinux
```

Patch it up

Apply the RTLlinux patch to the vanilla kernel source, using the following steps:

```
cd linux
patch -p1 < ../rtlinux/patches/kernel-patch-2.4.21-rtl3.2-pre3
```

You may see this error during patching:

```
patch: **** malformed patch at line 1047: */
```

If so, then edit `/rtlinux/patches/kernel_patch-2.4.21-rtl3.2-pre3` in your favourite editor. Go to line 1047 and change the following text:

```
(+/*
)
...to:
```

```
(+/*
)
```

Once again, extract the Linux source code (to revert to the original source) and then apply the modified patch, with the same commands given above.

Compile and install Linux

After the RTL patch is applied, proceed through the following steps to compile the kernel:

```
cd /usr/src/linux
make menuconfig
```

In the configuration menu, choose the exact processor type. You can find the name of the processor on your machine by running the `"cat /proc/cpuinfo"` command. Disable SMP

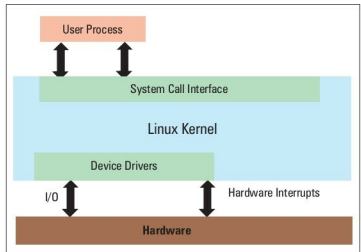


Figure 1: Linux kernel

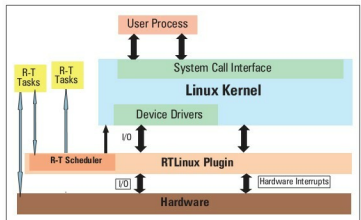


Figure 2: RTLlinux system

support, and set the version information on all module symbols. Then proceed with the following code:

```
make dep
make modules
make modules_install
make install
```

Reboot your system. In the GRUB menu, you will find a new entry for the newly compiled kernel.

Compiling and installing RTLlinux

Issue the following commands:

```
cd /usr/src/rtlinux
make menuconfig
make
make devices
make install
```

Post installation

After this, run the commands to check the status of the RTLlinux modules. First of all, you should run the regression test:

```
./scripts/regression.sh
```